

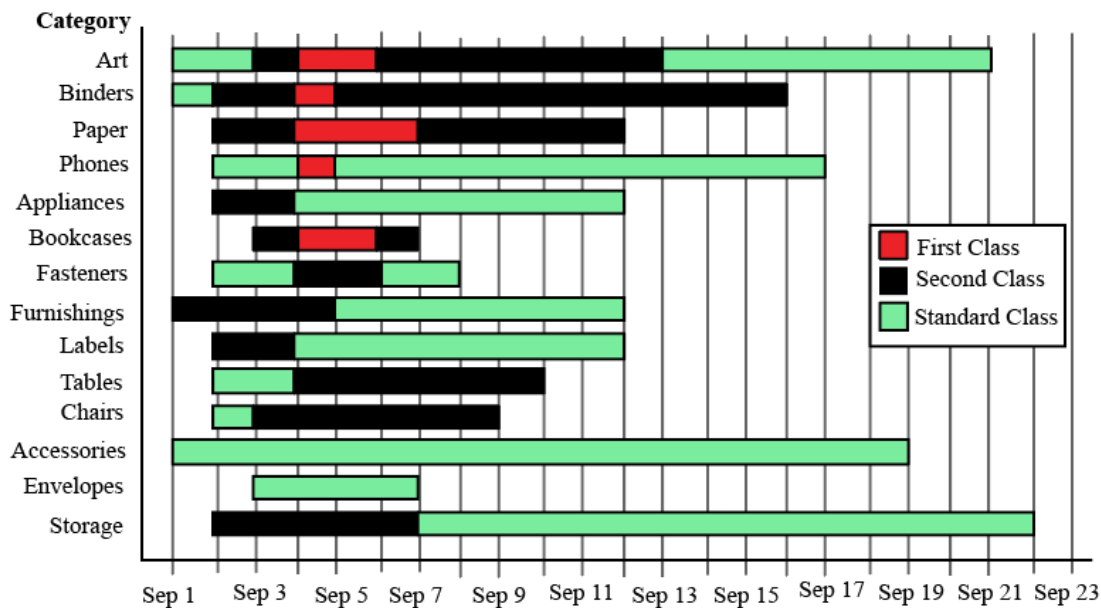


Top 12 CAT Data Interpretation Questions With Video Solutions

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Questions

Instructions



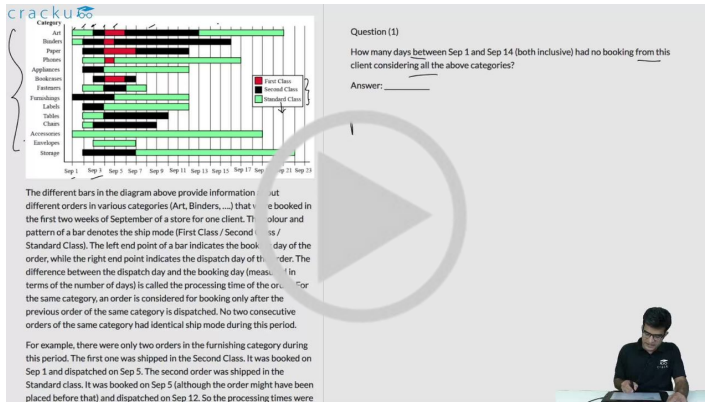
The different bars in the diagram above provide information about different orders in various categories (Art, Binders,) that were booked in the first two weeks of September of a store for one client. The colour and pattern of a bar denotes the ship mode (First Class / Second Class / Standard Class). The left end point of a bar indicates the booking day of the order, while the right end point indicates the dispatch day of the order. The difference between the dispatch day and the booking day (measured in terms of the number of days) is called the processing time of the order. For the same category, an order is considered for booking only after the previous order of the same category is dispatched. No two consecutive orders of the same category had identical ship mode during this period.

For example, there were only two orders in the furnishing category during this period. The first one was shipped in the Second Class. It was booked on Sep 1 and dispatched on Sep 5. The second order was shipped in the Standard class. It was booked on Sep 5 (although the order might have been placed before that) and dispatched on Sep 12. So the processing times were 4 and 7 days respectively for these orders.

Question 1

The sequence of categories -- Art, Binders, Paper and Phones -- in decreasing order of average processing time of their orders in this period is:

- A Art, Binders, Paper, Phones
- B Phones, Art, Binders, Paper
- C Phones, Binders, Art, Paper
- D Paper, Binders, Art, Phones



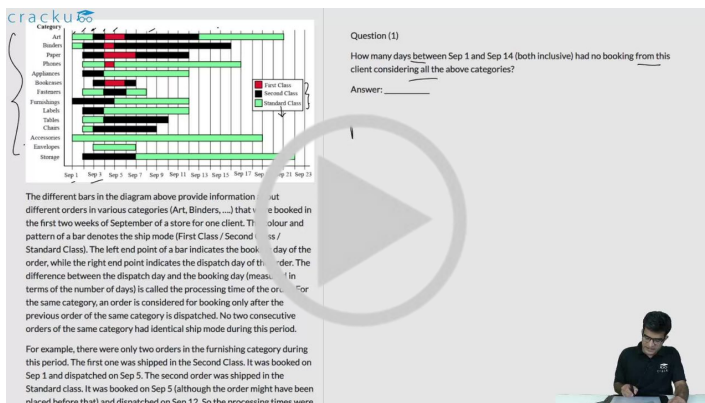
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Question 4

How many days between Sep 1 and Sep 14 (both inclusive) had no booking from this client considering all the above categories?

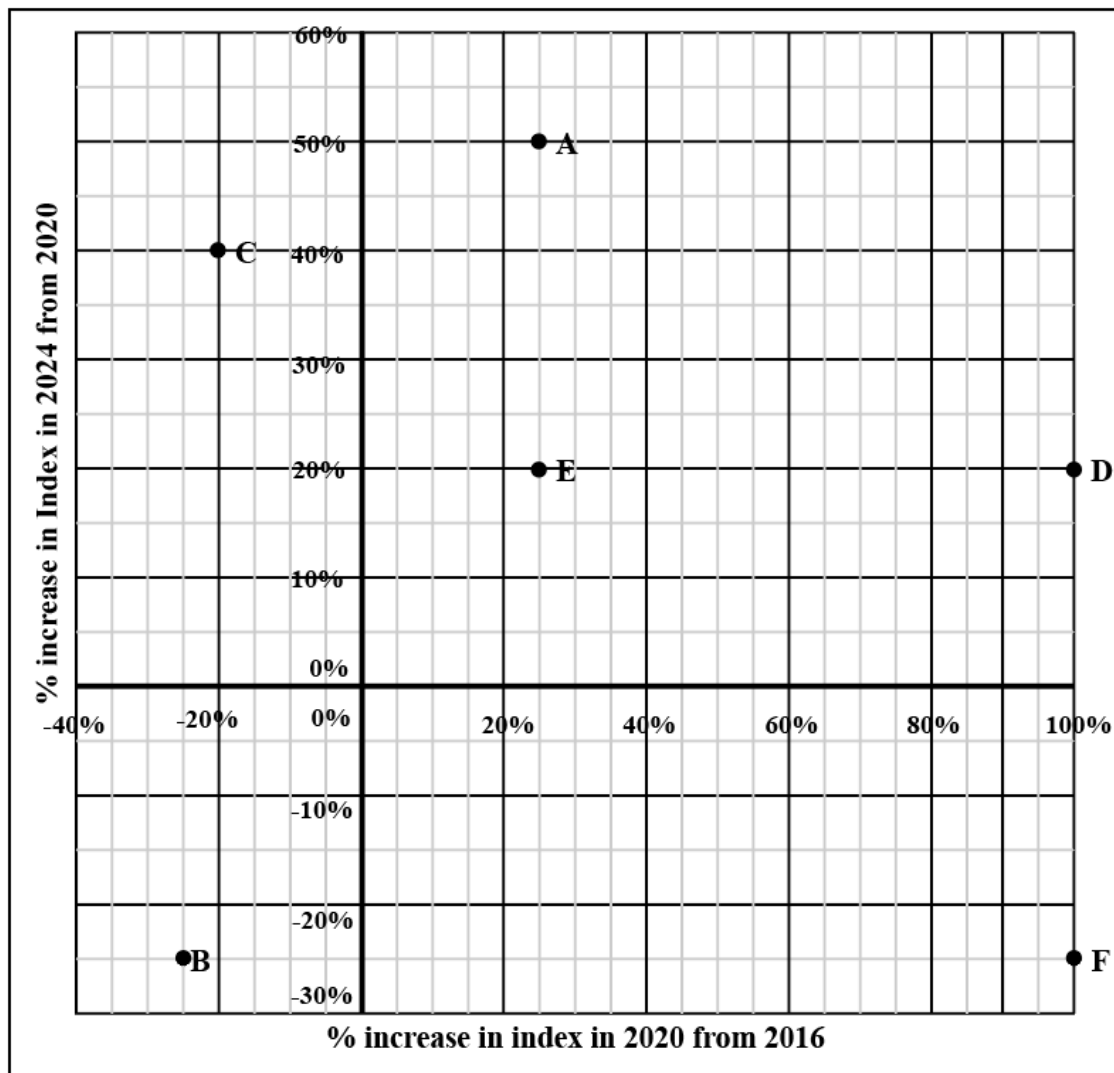


VIDEO SOLUTION

Instructions

The Sustainability Index (SI) of a country at a point in time is an integer between 1 and 100. This question is related to SI of six countries - A, B, C, D, E, and F - at three different points in time - 2016, 2020, and 2024. The plot represents the exact changes in their SI, with X-coordinate representing % increase in 2020 from 2016, i.e., $(SI \text{ in } 2020 \text{ minus } SI \text{ in } 2016) / (SI \text{ in } 2016)$, and Y-coordinate representing % increase in 2024 from 2020. At any point in time, the country with highest SI is ranked 1, while the country with the lowest SI is ranked 6. The following additional facts are known.

1. In 2016, B, C, E, and A had ranks 1, 2, 3, and 4 respectively.
2. F had lower SI than any other country in 2016, 2020, and 2024.
3. In 2024, E was the only country with SI of 90.
4. The range of SI of the six countries was 60 in 2016 as well as in 2024.



Question 5

What was the SI of C in 2024?

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Question 6

What was the SI of E in 2016?

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Question 7

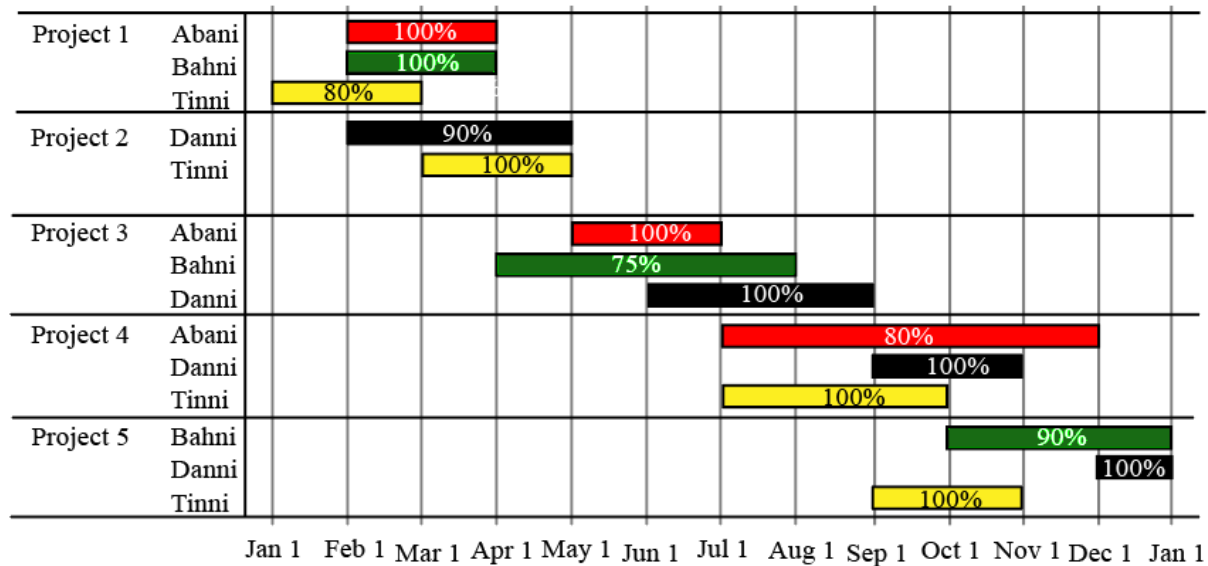
What was the SI of B in 2024?

- A 60
- B 54
- C 80
- D 45

Question 8

What was the SI of F in 2020?

Instructions



The figure above shows the schedule of four employees - Abani, Bahni, Danni, and Tinni - whom Dhoni supervised in 2020. Altogether there were five projects which started and concluded in 2020 in which they were involved. For each of these projects and for each employee, the starting day was at the beginning of a month and the concluding day was the end of a month, and these are indicated by the left and right end points of the corresponding horizontal bars. The number within each bar indicates the percentage of assigned work completed by the employee for that project, as assessed by Dhoni.

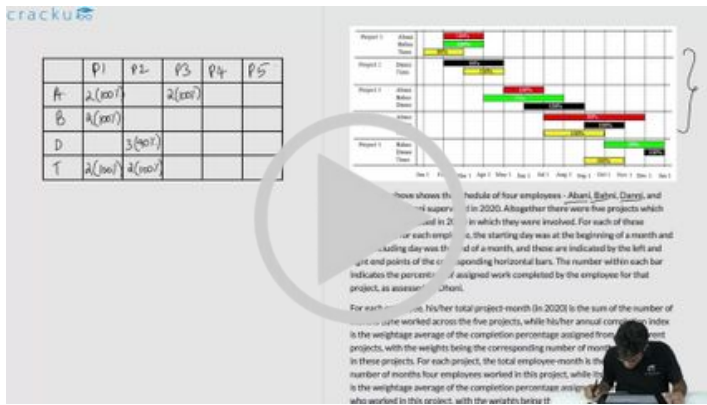
For each employee, his/her total project-month (in 2020) is the sum of the number of months (s)he worked across the five projects, while his/her annual completion index is the weightage average of the completion percentage assigned from the different projects, with the weights being the corresponding number of months (s)he worked in these projects. For each project, the total employee-month is the sum of the number of months four employees worked in this project, while its completion index is the weightage average of the completion percentage assigned for the employees who worked in this project, with the weights being the corresponding number of months they worked in this project.

Question 9

Which of the following statements is/are true?

- I: The total project-month was the same for the four employees.
 II: The total employee-month was the same for the five projects.

- A Only II
 B Both I and II
 C Neither I nor II
 D Only I



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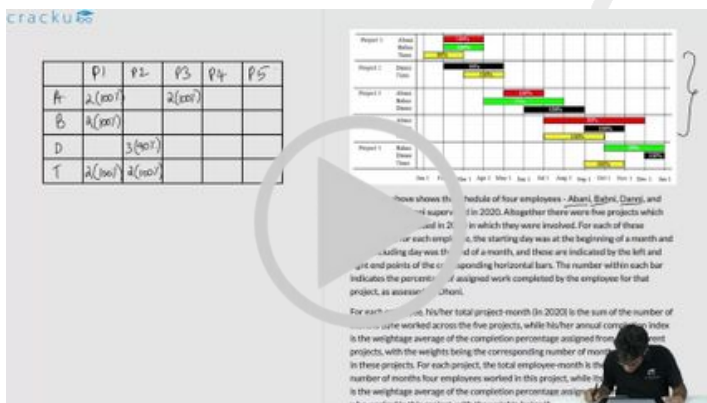
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Question 10

The project duration, measured in terms of the number of months, is the time during which at least one employee worked in the project. Which of the following pairs of the projects had the same duration?

- A Project 1, Project 5
- B Project 4, Project 5
- C Project 3, Project 5
- D Project 3, Project 4



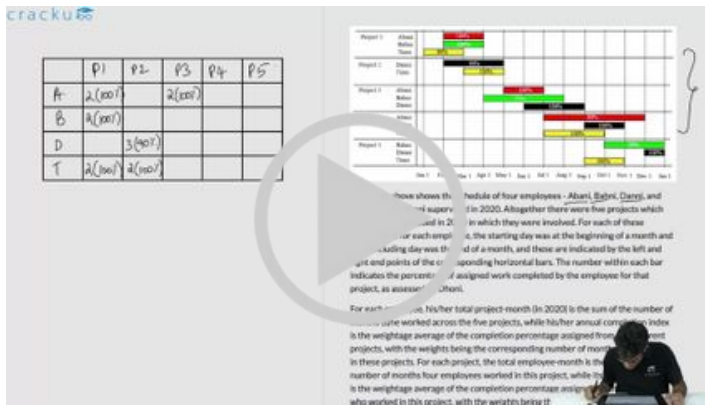
VIDEO SOLUTION

Question 11

Which employees did not work in multiple projects for any of the months in 2020?

- A Only Abani, Bahni and Danni
- B Only Abani and Bahni
- C All four of them

D Only Tinni

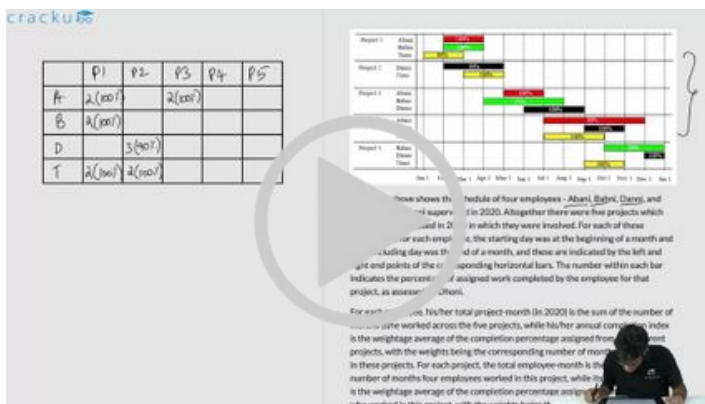


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Question 12

The list of employees in decreasing order of annual completion index is:

- A Danni, Tinni, Bahni, Abani
- B Bahni, Abani, Tinni, Danni
- C Danni, Tinni, Abani, Bahni
- D Tinni, Danni, Abani, Bahni



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CAT Syllabus PDF



Answers

1. **B** 2. **11** 3. **C** 4. **6** 5. **84** 6. **60** 7. **D** 8. **40**
9. **D** 10. **D** 11. **A** 12. **C**

Explanations

1. **B**

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 , 5-16
Paper	x	4-7	2-4 , 7-12
Phones	2-4 , 5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now taking average processing time per order for the above mentioned categories we get :

$$\text{Art} = \frac{2+8+2+1+7}{5} = 4$$

$$\text{Binders} = \frac{1+1+11+2}{4} = 3.75$$

$$\text{Papers} = \frac{3+2+5}{3} = 3.33$$

$$\text{Phones} = \frac{2+12+1}{3} = 5$$

So in decreasing order we get Phones , Art , Binder , Paper.

 VIDEO SOLUTION

2. **11**

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 ,5-16
Paper	x	4-7	2-4 ,7-12
Phones	2-4 ,5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now Envelopes and Accessories has only 1 ship mode i.e Standard class .

So therefore processing days for envelopes = 7-3 =4

and processing days for accessories = 19-1 =18

Therefore average = $\frac{(18+4)}{2} = 11$

 VIDEO SOLUTION

3.C

Accumulating all the data :

We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 ,5-16
Paper	x	4-7	2-4 ,7-12
Phones	2-4 ,5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now from the table we observe that the total number of orders are 35 and 7 orders have a processing time of 1 unit

The 7 orders are : Arts Standard class, Binders First class and standard class, Phones First class, Bookcases second class (2 orders) and Chairs standard class.

So the percentage = $\frac{7}{35} \times 100 = 20$

4.6

Accumulating all the data :
We get the following table :

Categories	Standard Class	First Class	Second Class
Arts	1-3 , 13-21	4-6	3-4 , 6-13
Binders	1-2	4-5	2-4 , 5-16
Paper	x	4-7	2-4 , 7-12
Phones	2-4 , 5-17	4-5	x
Appliances	4-12	x	2-4
Bookcases	x	4-6	3-4 , 6-7
Fasteners	2-4 , 6-8	x	4-6
Furnishings	5-12	x	1-5
Labels	4-12	x	1-3
Tables	2-4	x	4-10
Chairs	2-3	x	3-9
Accessories	1-19	x	x
Envelopes	3-7	x	x
Storage	7-22	x	2-7

Note a-b : represents the duration where a is the day when order is booked and b is the day when it is dispatched .

Now No booking days from the table are : September 8,9,10,11,12 and 14.
So a total of 6 days .

5.84

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{25}c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right) d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right) e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right) f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	5a/4		15a/8	
B	b	1	3b/4		9b/16	
C	c	2	4c/5		28c/25	
D	d	5	2d		12d/5	
E	e = 60	3	5e/4 = 75		3e/2 = 90	
F	f	6	2f	6	3f/2	6

We know that F is ranked 6 and B is ranked 1 in 2016, and we are also given that the range of SI in 2016 and 2024 is 60. This means that the value of b-f = 60, and we can conclude that b is definitely greater than 60, as we know that f is an integer greater than 0.

We know that all the SI values in all the years are integers, and for SI of B to be an integer in 2024, the value of b must be a multiple of 16. The multiples of 16 that are greater than 60 and less than 100 are 64, 80 and 96.

CASE 1: b = 64

If $b = 64$, then the value of f becomes 4, as we know $b - f = 60$.

If $f = 4$, then the value of F in 2024 is $3f/2 = 12/2 = 6$.

We already know the value of E in 2024 is 90.

In that case, the range of SI in 2024 is at least $90 - 6 = 84$, but we are given that the range is 60 in 2024, which is not possible in this case.

So, we can eliminate this case.

CASE 2: $b = 96$

If $b = 96$, then the value of f becomes 36, as we know $b - f = 60$.

If $f = 36$, then the value of F in 2024 is $3f/2 = 108/2 = 54$.

The value of B in 2024 = $9b/16 = (9 * 96)/16 = 54$

We calculated the value of B and F to be 54 in 2024, which is not possible, as we are given that F had a lower SI than any other country in 2024, and we obtained its value to be the same as B .

So, we can eliminate this case.

CASE 3: $b = 80$

If $b = 80$, then the value of f becomes 20, as we know $b - f = 60$.

If $f = 20$, then the value of F in 2024 is $3f/2 = 60/2 = 30$.

The value of B in 2024 = $9b/16 = (9 * 80)/16 = 45$

We can see that no condition is violated in this case, as F is less than B in 2024, and the range in 2024 from the obtained values as of now is $90 - 30 = 60$.

So, we can conclude that the value of $b = 80$ and $f = 20$.

Substituting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	$b = 80$	1	$3b/4 = 60$		$9b/16 = 45$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	$e = 60$	3	$5e/4 = 75$		$3e/2 = 90$	1
F	$f = 20$	6	$2f = 40$	6	$3f/2 = 30$	6

We know that C is ranked 2nd in 2016, and we know the values of 1st and 3rd to be 80 and 60. We also know that SI of C has to be integers in 2016, 2020 and 2024. So, for the value of SI of C to be an integer in 2024, the value of c has to be a multiple of 25.

So, we know that c lies between 80 and 60, and is also a multiple of 25. The only multiple of 25 that lies between 60 and 80 is 75. So, the value of c has to be 75, and the value of C in 2024 can be calculated as,

C in 2024 = $28c/25 = (28 * 75)/25 = 84$.

Hence, the correct answer is 84.



CAT Formula PDF

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6.60

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

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For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)d = 2d$$

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For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	b	1	$3b/4$		$9b/16$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	e = 60	3	$5e/4 = 75$		$3e/2 = 90$	
F	f	6	$2f$	6	$3f/2$	6

SI of E in 2016 is 60.

Hence, the correct answer is 60.

[VIEW SOLUTION](#)

7. D

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

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For C:

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We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

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We know that all the SI values in all the years are integers, and for SI of B to be an integer in 2024, the value of b must be a multiple of 16. The multiples of 16 that are greater than 60 and less than 100 are 64, 80 and 96.

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We already know the value of E in 2024 is 90.

In that case, the range of SI in 2024 is at least $90 - 6 = 84$, but we are given that the range is 60 in 2024, which is not possible in this case.

So, we can eliminate this case.

CASE 2: $b = 96$

If $b = 96$, then the value of f becomes 36, as we know $b - f = 60$.

If $f = 36$, then the value of F in 2024 is $3f/2 = 108/2 = 54$.

The value of B in 2024 = $9b/16 = (9 * 96)/16 = 54$

We calculated the value of B and F to be 54 in 2024, which is not possible, as we are given that F had a lower SI than any other country in 2024, and we obtained its value to be the same as B .

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If $b = 80$, then the value of f becomes 20, as we know $b - f = 60$.

If $f = 20$, then the value of F in 2024 is $3f/2 = 60/2 = 30$.

The value of B in 2024 = $9b/16 = (9 * 80)/16 = 45$

We can see that no condition is violated in this case, as F is less than B in 2024, and the range in 2024 from the obtained values as of now is $90 - 30 = 60$.

So, we can conclude that the value of $b = 80$ and $f = 20$.

Substituting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	$b = 80$	1	$3b/4 = 60$		$9b/16 = 45$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	$e = 60$	3	$5e/4 = 75$		$3e/2 = 90$	1
F	$f = 20$	6	$2f = 40$	6	$3f/2 = 30$	6

So, SI of B in 2024 is 45.

Hence, the correct answer is option D.

[VIEW SOLUTION](#)

8.40

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A :

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{25}c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	b	1	$3b/4$		$9b/16$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	$e = 60$	3	$5e/4 = 75$		$3e/2 = 90$	
F	f	6	$2f$	6	$3f/2$	6

We know that F is ranked 6 and B is ranked 1 in 2016, and we are also given that the range of SI in 2016 and 2024 is 60. This means that the value of $b - f = 60$, and we can conclude that b is definitely greater than 60, as we know that f is an integer greater than 0.

We know that all the SI values in all the years are integers, and for SI of B to be an integer in 2024, the value of b must be a multiple of 16. The multiples of 16 that are greater than 60 and less than 100 are 64, 80 and 96.

CASE 1: $b = 64$

If $b = 64$, then the value of f becomes 4, as we know $b - f = 60$.

If $f = 4$, then the value of F in 2024 is $3f/2 = 12/2 = 6$.

We already know the value of E in 2024 is 90.

In that case, the range of SI in 2024 is at least $90 - 6 = 84$, but we are given that the range is 60 in 2024, which is not possible in this case.

So, we can eliminate this case.

CASE 2: $b = 96$

If $b = 96$, then the value of f becomes 36, as we know $b - f = 60$.

If $f = 36$, then the value of F in 2024 is $3f/2 = 108/2 = 54$.

The value of B in 2024 = $9b/16 = (9 * 96)/16 = 54$

We calculated the value of B and F to be 54 in 2024, which is not possible, as we are given that F had a lower SI than any other country in 2024, and we obtained its value to be the same as B.

So, we can eliminate this case.

CASE 3: $b = 80$

If $b = 80$, then the value of f becomes 20, as we know $b - f = 60$.

If $f = 20$, then the value of F in 2024 is $3f/2 = 60/2 = 30$.

The value of B in 2024 = $9b/16 = (9 * 80)/16 = 45$

We can see that no condition is violated in this case, as F is less than B in 2024, and the range in 2024 from the obtained values as of now is $90 - 30 = 60$.

So, we can conclude that the value of $b = 80$ and $f = 20$.

Substituting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	$b = 80$	1	$3b/4 = 60$		$9b/16 = 45$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	$e = 60$	3	$5e/4 = 75$		$3e/2 = 90$	1
F	$f = 20$	6	$2f = 40$	6	$3f/2 = 30$	6

So, SI of F in 2020 is 40.

Hence, the correct answer is 40.

[VIEW SOLUTION](#)

9. D

The total project month is the number of months Abani, Bahni, Danni, and Tinni individually worked for all the projects combined :

Abani - $2+2+5 = 9$ months

Bahni - $2+4+3 = 9$ months

Danni - $3+3+2+1 = 9$ months

Tinni - $2+2+3+2 = 9$ months.

The total employee month for all the five projects is the sum of the total employee-month is the sum of the number of months four employees worked in this project.

Project -1 = $2+2+2 = 6$ months

Project -2 = $3+2 = 5$ months

Project - 3 = $2+4+3 = 9$ months.

Project - 4 = $5+2+3 = 10$ months.

Project - 5 = $3+1+2 = 6$ months.

Only statement 1 is true.

[VIDEO SOLUTION](#)

10. D

Considering the information provided :

For project 1 : 3 months.

Project - 2: 3 months.

Project - 3: 5 months.

Project - 4: 5 months.

Project - 5: 4 months.

Among the given options option D is true which is project 3, project 4.

 VIDEO SOLUTION



11. A

Abani, Banni, and Danni did not work on multiple projects simultaneously in a month

Tinni was the only person who worked on multiple projects which are project 4 and project 5 in the month of september.

 VIDEO SOLUTION

12. C

The annual completion index for different people is :

The weightage average of the completion percentage assigned from the different projects, with the weights being the corresponding number of months (s)he worked in these projects.

For Abani :

$$\frac{((100 \cdot 2) + (100 \cdot 2) + (80 \cdot 5))}{2 + 2 + 5} = \frac{800}{9}$$

For Bahni :

$$\frac{((100 \cdot 2) + (75 \cdot 4) + (90 \cdot 3))}{2 + 3 + 4} = \frac{770}{9}$$

For Danni :

$$\frac{((90 \cdot 3) + (100 \cdot 3) + (100 \cdot 2) + (100 \cdot 1))}{2 + 3 + 2 + 1} = \frac{870}{9}$$

For Tinni :

$$\frac{((80 \cdot 2) + (100 \cdot 2) + (100 \cdot 3) + (100 \cdot 2))}{2 + 2 + 3 + 2} = \frac{860}{9}$$

The descending order for the four people is :

Danni, Tinni, Abani, Bahni.

 VIDEO SOLUTION